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FIRE Rule From a Perspective of a Czech Retail Investor

Bachelor's thesis

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Declaration of Authorship

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Abstract

The thesis estimates a safe withdrawal rate for a Czech retail investor interested in the Financial Independence, Retire Early (FIRE) movement. It is assumed the investor aims for a 30–year payout period and their portfolio is active in up to three markets: the Czech Republic, the Eurozone, and the US. The market considerations are divided into four scenarios, representing different allocations of funds across the markets. Furthermore, the thesis assumes a fixed construction of the portfolio in assets to be an equal combination of stocks and bonds. The estimation is approached through a Monte Carlo Simulation that generates annual logarithmic returns of the assets and the exchange rates when applicable, while employing Cholesky decomposition to capture the empirical correlation among the variables. Inflation is modelled independently, and different treatments are applied depending on whether the investor owns a property. In total, seven different withdrawal rates are investigated, and it is concluded the safe withdrawal rate, or the FIRE rule, is 2.5% for the Czech investor in most of the scenarios considered, but can rise to 3% with international diversification and own housing.

JEL Classification	C15, C63, G11, G15, G17
Keywords	Financial Independence—Retire Early, FIRE, Czech Retail Investor, Safe Withdrawal Rate, Investment Portfolio, International Diversification, Monte Carlo Simulation
Title	FIRE Rule From a Perspective of a Czech Retail Investor

Abstrakt

Tato práce odhaduje bezpečnou míru výběru z investičního portfolio pro Českého malého investora, který se zajímá o hnutí Financial Independence, Retire Early (FIRE). Je předpokládáno, že investor míří na 30letou dobu výběrů a s daným portfoliem je aktivní na až třech trzích: Českém, Americkém a v Eurozóně. Role aktivity na trzích je rozdělena do čtyřech případů, které představují různé alokace prostředků mezi tyto trhy. Dále tato práce předpokládá pevnou konstrukci portfolia v rámci rozdělení mezi akcie a dluhopisy stejným dílem. K odhadu je přistupváno prostřednictvím simulace Monte Carlo, která

generuje roční logaritmické výnosy prostředků a případně směnných kurzů, přičemž pro zachycení empirické korelace mezi jednotlivými proměnnými se používá Choleského rozklad. Inflace je modelována nezávisle a v závislosti, zda investor vlastní nemovitost, jsou uplatňovány různé postupy. Celkem je zkoumáno sedm různých měr výběru a dochází se k závěru, že bezpečná roční míra výběru, tedy FIRE Rule, je 2.5% pro Českého investora ve většině zkoumaných případů a 3% pouze při aktivitě na všech trzích a s vlastním bydlením.

Klasifikace JEL	C15, C63, G11, G15, G17
Klíčová slova	Financial Independence—Retire Early, FIRE, Český malý investor, Bezpečná míra výběru, Investiční portfolio, Mezinárodní diverzifikace, Monte Carlo simulace
Název práce	FIRE Rule z Pohledu Českého Malého Investora

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Acronyms

CZK Czech Koruna

EUR Euro

EU European Union

FIRE Financial Independence, Retire Early

MCS Monte Carlo Simulation

RIC Reuters Instrument Code

SWR Safe Withdrawal Rate

USD United States Dollar

US United States of America

Chapter 1

Introduction

Planning for retirement is inevitable for every labour force participant, but the anticipation can seem endless for some. The movement behind shortening the waiting is called Financial Independence, Retire Early (FIRE). Its emphasis on high savings, complemented with minimalism and financial literacy, aims to retire sooner than is the government–set age (Taylor & Davies, 2021). The adherents then enjoy the lack of necessity to work, which comes in the form of free time. Motivations behind achieving financial independence may vary, but studies identified that having more time to pursue either personal hobbies and well–being or volunteering plays a significant role (Perrone *et al.*, 2015; Dittmar *et al.*, 2014). In order to accumulate sufficient funds to retire, a member should know the rate at which they can withdraw safely so as not to deplete the principal prematurely.

The popular approach in the community to spending is the FIRE Rule, also called the 4% Rule. The rule was first estimated by Bengen (1994). The author showed, using historical backtesting, that a retiree with a portfolio constructed of an equal share of stocks and bonds can spend 4% of the initial portfolio value, adjusting it by inflation, when the desired payout period is 30 years. On the other hand, the validity of the rule has not been confirmed by other researchers. Although some studies indicate the investor can safely indulge even higher withdrawals (e.g., Finke *et al.*, 2012; Duquette, 2023), others, especially those investigating the rule in different countries, argue that the rule is overestimated (e.g., Pfau, 2014; Tamimi *et al.*, 2024).

Most of the literature regarding the FIRE rule is focused on one market only. However, portfolio performance may also be enhanced by international diversification (e.g., Liu, 2016; Attig *et al.*, 2023; Levy & Lerman, 1988). That would

argue for higher withdrawals. It is now fairly easy to invest internationally with online brokers, and the literature regarding the FIRE rule under an internationally diversified portfolio is not extensive. In other words, many studies are showing that diversification has a positive effect on portfolio performance (e.g., Liu, 2016; Attig *et al.*, 2023; De Santis & Sarno, 2008), but the effect's translation to the safe withdrawal rate has not been thoroughly researched.

The primary purpose of this thesis is to obtain the FIRE rule for a Czech retail investor. The characteristics of an investor based in the Czech Republic and in the United States of America (US) as well as the respective markets differ, and therefore, the rule may differ too. Besides the evident differences in markets and therefore different returns on the instruments, the investors also differ in housing. Most FIRE rules are calculated under the assumption that the rentier lives in a rental. However, Czechs are likely to own a property (Statista, 2025). The question then is whether the ownership of real estate has a significant effect on the withdrawal rate. The thesis also allows for an international construction of an investment portfolio.

It is assumed a retiree's goal is a 30–year payout period with fixed construction of a portfolio in 50% stocks and 50% bonds. The assets are then taken from three markets that are assumed to be interesting for a Czech retail investor: the Czech, US, and European, or more precisely, Eurozone, markets. Then, four different splits across the markets are investigated. One that considers only domestic assets, another one with only foreign assets, and two showing some combination of all three. Withdrawals as low as 2% and as high as 5% are considered for the computation.

The estimation of the FIRE rule is approached through a Monte Carlo Simulation method that mimics the behaviour of the assets that the given portfolio is constructed of. The simulation generates yearly correlated returns for all of the assets that the portfolio is constructed of, and the exchange rate, if present. Inflation is generated independently. The appropriate amount is then withdrawn, and the portfolio is rebalanced on instruments. For every portfolio (run), the simulation continues for 30 years or until the portfolio is exhausted, whichever happens first. Finally, the number of portfolios outlasting the payout period over the number of runs is reported as the survival rate. A withdrawal rate is considered safe if the portfolio survives at least 95% of the cases.

The thesis has the following structure. Chapter 2 provides thorough insight into the background of the research problem. Chapter 3 describes the methodology used to address the estimation, followed by a description and a

visualisation of the data used in the analysis illustrated in Chapter 4. The findings are reported in Chapter 5. Chapter 6 provides a comprehensive summary and Appendix A shows the average performance of every Monte Carlo Simulation.

Chapter 2

Literature Review

The literature review covers relevant research for investigating the 4% Rule to date. It is sectioned into three parts. The first part covers the movement behind the rule. It goes through the motivations adherents have to reach financial independence and their practices to achieve the goal or the definition of financial independence in the community. Then, it is followed by a section about the 4% Rule itself. It discusses its origins, limitations and international applicability. The last point is just as important because it hints at whether the rule is likely the same or higher/lower for a retail investor in the Czech Republic. Since the investment portfolio constructed for the analysis considers activity in three markets, the chapter is closed with a section about diversification. It addresses the importance of diversification and concludes its benefits and drawbacks.

2.1 FIRE Movement

Studies indicate that the Financial Independence, Retire Early movement first emerged in the 1990s in the United States. What initially started as a collection of seminars and educational materials about early retirement by Dominguez and Robin (whose book *Your Money or Your Life* has been the main source of the philosophy) continued to be an inspiration to others, further invoking research on finding the best approach to achieve the goal (Avendaño Miranda, 2024; Taylor & Davies, 2021).

FIRE can be characterised in the following terms. Firstly, it emphasises the need to reduce consumption and increase savings while also avoiding debt. Many members practice minimalism to achieve this goal. Another important

characteristic is that it relies on passive investment strategies based on index funds rather than using portfolio managers or trading. Last but not least, it is an economic philosophy that values free time more than financial wealth. For many retirees, the absence of an obligation to work is the most satisfactory. As Robin *et al.* (2008) put it: 'You pay for money with your time'.

In the community, financial independence is defined as the freedom from necessity. The society sees power in self–help which is projected in the absence of portfolio managers, as well as minimalism that helps with lessening living expenses. It is believed that frugality evoking independence from both the labour market and consumerism brings freedom. And while minimalism can help greatly with lowering one's expenses, budgeting, as well as investments in financial instruments, is potentially even more important. Hence, financial literacy is necessary when achieving FIRE. Taking this into consideration, it is not surprising that successful participants often come from fields similar to software engineering and finance, where high income combined with low expenditures allow for substantial savings (Taylor & Davies, 2021).

Although every adherent wants to achieve *financial independence*, the motivations behind achieving it, how the allocated free time is spent, and the savings pace can differ greatly within the group. The common identifiers among the early retirees are volunteerism–motivated goals. The goals take the form of 'making the world a better place' or 'helping others'. Devoting more time to own hobbies and focusing on one's well–being also appears among the motivations (Perrone *et al.*, 2015).

Regardless of the motivation, three main techniques to achieving financial independence exist: *Lean FIRE*, *FIRE*, and *Fat FIRE*. Some define individual approaches in terms of the percentage saved relative to average household income, while others define them more loosely. Nevertheless, *Lean FIRE* is the approach that rushes to retirement the fastest. Individuals save aggressively, practice minimalism and retire with comparably low income, but sooner than the rest. Traditional *FIRE* then refers to those who save enough that they are able to withdraw at least the median income from their investment portfolio. These adherents likely retire later than *Lean FIRE* practitioners, however, their income is also higher. The third category, *Fat FIRE*, on the other hand, does not rush into retirement. Although the adherents also save a substantial part of their income, they work long enough not to have to lower their standard of living (Adcock, 2024; Mitra & Barba; 2024). In other words, the sooner the investor aims to retire, the lower is their required cost of living.

This thesis does not differentiate between types of FIRE since it does not matter what the investment portfolio's absolute value is to determine the ideal initial withdrawal rate. In other words, it does not examine whether the withdrawn amount is *enough* to fund an individual's expenses, it only determines the share withdrawn in terms of the portfolio.

Overall, FIRE as a movement does not have one strict definition that people would have to follow. Instead, it has several approaches that followers take based on their individual preferences and needs. However, they all share one feature: the necessity of sufficient funds to live on. These funds are typically stored in an investment portfolio to fight against inflation and gain some additional earnings. How much individuals should withdraw not to deplete their savings prematurely, while also not leaving much left after the payout period, is explored in the following Section 2.2.

2.2 4% Rule

Arguably, the paper that has influenced the movement's members the most is Bengen's *Determining Withdrawal Rates Using Historical Data* (1994) (Milevsky & Huang, 2011). Bengen (1994) examines the longevity of different withdrawal rates in combination with different stock/bond allocation portfolios and comes to a conclusion, which is now considered a rule of thumb (Scott *et al.*, 2008). For a portfolio constructed of 50% stocks and equally 50% bonds, the ideal initial withdrawal rate is 4% when the desired longevity is 30 years. Bengen (2006) further increased the rate to 4.5%, however, most of the subsequent literature refers to the original 4% Rule (e.g., Scott *et al.*, 2008; Clare *et al.*, 2017; Costa *et al.*, 2021). Accordingly, this thesis adopts the conventional 4% rule as its primary benchmark.

For a better explanation of the withdrawal strategy, consider the following example. An investment portfolio is valued at \$1,000,000 at the beginning of the payout period. The retiree hence withdraws \$40,000, corresponding to 4%, at that year. Without loss of generality, assume the inflation between years 1 and 2 is 2%. Then, the withdrawal at year 2 is $\$40,000 \cdot 1.02 = \$40,800$, no matter the current value of the portfolio.

The advantage of this method is its easy application, allowing for the reduction of costs, e.g., in the absence of a portfolio manager. On the contrary, this approach faces difficulties when reacting to shocks in the portfolio value, i.e., shocks in the underlying resources. Since the retiree withdraws the same

real amount every year, the percentage withdrawn can significantly drift from the initial 4% in either direction, depending on whether the instruments are under or overperforming.

Another widely cited study in the community is the so–called Trinity Study by Cooley *et al.* (1998). The authors conclude that the chance of a portfolio lasting 30 years when the initial withdrawal rate is set to 4% and the stock/bond rate is 50/50 is about 95%, supporting Bengen's findings. It is worth noting that they also consider that many retirees want to leave something to their heirs, which would argue for a lower withdrawal rate. This analysis assumes a selfish retiree who would rather maximise personal consumption than leave an inheritance.

Apart from the 30–year period, the authors consider withdrawal periods to be as low as 15 years. The length of a payout period varies across individuals and relates to the approach to FIRE taken. Since the average life expectancy was 79 years in 2022 in the Czech Republic and the government–set retirement age was 65 years of age at the time, even a 15–year–long retirement could be reasonable (World Bank Group, 2022; Czech Social Security Administration, 2022). Nonetheless, the goal of FIRE is to *retire early*. Hence, the 30–year or longer payout periods seem more reasonable.

According to some, the 4% withdrawal rate may be too low or optimal for risk–averse clients only (Finke *et al.*, 2012; Kitces, 2008). The general understanding is that higher withdrawal rates can be afforded mainly when equity is dominating in the portfolio (e.g. 60% or 70% in stocks) (e.g. Bengen, 1994; Cooley *et al.*, 1998). Duquette (2023) finds that the SAFEMAX initial withdrawal rate, which is a term Bengen (1994) coined for the highest portfolio withdrawal rate adjusted for inflation that would last at least 30 years, can be even close to 5% when the equity is more diversified and includes some US mid–cap stocks.

Even though the 4% Rule is popular in the FIRE community, its perfection has been questioned by academics and concerns regarding the optimality of the approach have been raised by several. Milevsky & Huang (2011) criticise the lack of background in economic theory and develop a model regarding the problem. They include 'longevity risk' (risk of outliving the payout period) and conclude that though the 4% rule strategy may be optimal for some retirees, it is not the best for most and rather other withdrawal approaches should be considered. They also criticise the lack of adaptability to shocks. Scott *et al.* (2008) focus on over–payments the rule of thumb brings. Again, this is based

on the approach's constant, non–volatile nature, and they conclude that 10%–20% of a portfolio's initial wealth is allocated to surpluses and an additional 2%–4% to overpayments. Furthermore, its relevance to date is not guaranteed.

4% Rule was estimated more than 30 years ago, and while it was constructed to be applicable in the long term, it is worth examining the current validity of the approach. Tamimi *et al.* (2024) conducted a Monte Carlo Simulation with random shock (representing, e.g., the 2008 Global Financial Crisis or the COVID-19 pandemic) on a US portfolio with annually–inflation–adjusted 4% withdrawal rate. With a 50% stocks, 50% bonds structure of the portfolio and two market crashes in the payout period, the probability of depleting the portfolio prematurely was 22% when the portfolio's value declined by 5%, and 33% when it declined by 15%. With three market crashes, the probability of a failure was 27% and 46%, respectively. Hence, the approach can be relatively risky, at least under the last scenario. Though more stocks–based portfolios perform better in all mentioned cases, the chance of a default is still 38% with a 70/30 stock/bond allocation and three market crashes. The overall results then suggest that while Bengen's approach may work in most cases, it does carry a previously nonvocalized risk.

Pfau (2010) examined the withdrawal rate using Bengen's methodology from an international perspective, considering 17 countries over 109 years. While applying the SAFEMAX criterion and using an even allocation of stocks and bonds, the 4% Rule fails in all examined countries. The highest safe withdrawal rate was Canada's 3.94%. This could be caused, e.g., by the assumption that the local stock market will provide a positive real return over a 20–year period, which, however, holds for only four countries in the sample (Dimson *et al.*, 2003). Though this is true when the portfolio is constructed of financial assets in a single country, it may not be so with international diversification. The sample does not contain the Czech Republic.

Alternative approaches evolved from the 4% rule, trying to improve the approach. One of them is Vanguard's dynamic approach to retirement spending, which introduces both a floor and a ceiling to annual withdrawals. This approach then better reacts to shock in the markets with additional savings in times of the market's over-performance and reducing spending in times of its under-performance while allowing the retiree to rely on a quasi–fixed budget (Costa *et al.*, 2021). Another approach modifying Bengen's rule is to withdraw given 4% every year as a share of the portfolio's current value. This method, however, only increases the volatility in withdrawals and may be sub–optimal

for retirees since their living standards could vary greatly throughout the payout years.

While the FIRE rule has limitations and cannot be considered perfect, it does attract retirees' attention, perhaps due to its simple and direct approach. Its application in the Czech context is investigated in the thesis. Most of the examined papers focus on performance in one market only, yet this text considers activity in three markets simultaneously. For that reason, diversification is discussed in Section 2.3.

2.3 Diversification

While the specific impact of international diversification on the investment portfolio of a Czech retail investor has not been extensively studied, it can be presumed that it is consistent with the influence in other countries, for which the literature is extensive. This section reviews the research and examines the implications of international structure for the portfolio.

Liu (2016), again from the perspective of a US investor, finds that bond diversification can significantly reduce volatility and increase risk-adjusted returns. Viciera & Wang (2018) found a negative correlation between stocks and bonds in the US both within and across countries, showing that the benefits of diversification have increased in developed markets in recent years. Another paper by Eun & Resnick (1994) examines the benefits from US and Japanese perspectives and find that, though investors from both countries can benefit from diversification, the benefit itself varies. Investors based in the United States benefit from higher returns, while Japanese investors may enjoy the benefits of less risk. This supports the argument that the advantage of diversification is not universal, and one cannot expect to experience the same gains as another investor located in a different country or even a different continent.

De Santis & Sarno (2008) investigated the benefits of diversification in, at the time, current period when markets are increasingly more financially integrated, an effect that presumably will continue. While some countries' assets do perfectly comove with US ones over the studied period of 1991–2007, which would provide little benefit for the portfolio if used as a diversification tool, other countries do not possess this, in this case, unfortunate attribute and investing in them is beneficial.

The benefits of diversification, even with growing globalisation and global crises, are confirmed by Attig *et al.* (2023). The authors examined a period

of 1995–2021 (which includes e.g. IT bubble burst, global financial crisis and COVID–19) and found out that international diversification even dominates industrial diversification. The advantage international diversification provides is driven by the alleviation of market, political, and inflation risks. On the contrary, the risk faced by the quality of the legal and credit environment appears less important.

Most research on diversification is focused on the performance of international stocks. However, the portfolio considered in this thesis is also constructed of 50% bonds. Levy & Lerman (1988) investigated the benefits of investing in bonds internationally from the perspective of a US investor over a 20–year period, 1960–1980. In the period, not only was it beneficial to diversify bond investments internationally, but the benefits also outperformed those of an internationally diversified stock portfolio. The reason is the low correlation across world bond markets compared to the one in world stock markets. Nonetheless, the combination of diversification in both instruments performed the best.

With its application in several countries worldwide, Bengen's established approach did not prove to be perfect. Pfau (2014) finds a SAFEMAX rate of only 0.25% for a Japanese investor with an even stock–bond allocation in the local market and 30–year payout period. Moreover, Norway's failure rate is 53.6% in a similar setting. The author thus examines the benefits of international diversification on SAFEMAX rates in terms of a domestic currency. Although global diversification did not improve the rates for all 20 countries in the considered sample, it did improve them in 15 of them, and the other five countries had consistently strong–performing domestic markets. Overall, despite the fact that global diversification is not 'a universal panacea,' it did improve the aggregated success rates of a 4% initial withdrawal rate on a 50% stocks, 50% bonds portfolio from 65.7% with domestic assets to 78.3% in global assets.

Even though the Czech Republic is not frequently picked for the discussion of benefits of international diversification, Avdulaj & Barunik (2013) examined the benefits in the period 2008–2013 (i.e. up to 5 years after the global financial crisis) and used Czech PX and German DAX broad stock indices as diversification tools. The two indices are highly correlated, and even so, the authors found benefits in diversification between them. Nevertheless, the benefits vary strongly over time and are believed to disappear when the Czech Republic adopts the euro due to the strong connection between the economies.

When the portfolio is internationally diversified, the returns in foreign coun-

tries are not immediately transmitted to the domestic portfolio. An exchange rate between the currencies plays a vital role since the value of foreign assets increases with the devaluation of the local currency and vice versa. In this thesis, Czech Koruna (CZK) is the domestic currency, and United States Dollar (USD) and Euro (EUR) are treated as foreign currencies.

To provide an example, let a Czech retail investor have a USD1,000 in US stocks. At day 1, when the exchange rate is CZK24.00 for USD1, the stocks represent CZK24,000 for the investor. However, on day 2, when the exchange rate is CZK23.89 for USD1 (hence the koruna appreciated against the dollar), the investor values the stocks at only CZK23,890, ceteris paribus.

As valid a concern as exchange currency risk is, some studies show it can be mitigated in the long run. Fooladi & Rumsey (2006) show that the benefits of international diversification persist, with a possible explanation being that the movements in currency markets offset the movements in equity markets measured in local currency. On the other hand, Sgammini & Muzindutsi (2020) explore the matter from the perspective of a South African retail investor and comes to the conclusion that exchange rate shocks have a lasting effect on an investment portfolio's returns. The authors also argue that diversification among various foreign countries and various currencies helps mitigate the exchange rate risk.

Ziobrowski & Ziobrowski (1995) show that the exchange rate risk plays such a role that the optimum level varies directly with the investor's willingness to accept risk. Their study claims that the benefits of international diversification are limited by the exchange rate risk, which is consistent with later research. Apart from that, the study also suggests that a common stock may be a sensible foreign asset to hold since the acquisition of a single asset generates the majority of gains available in a foreign country. Further gains from the possession of additional assets are minimal due to the high positive correlation among asset returns in a given country.

While in practice, there are several methods used for hedging the exchange rate risk (e.g., Bae *et al.*, 2018; Yu *et al.*, 2020), this thesis does not treat the risk in any way. Be that as it may, the exchange rate is incorporated into the modelling and does affect the outcomes.

The last part of Chapter 2 discussed the impact of diversification on an investment portfolio. While the impact is softened by exchange rate risk, there is strong evidence it benefits the given account, even in the increasingly connected world. Whether the advantage for a Czech retail investor is in terms of

higher returns or lower risk is not clear, but if there is either of them, it likely causes the Safe Withdrawal Rate (SWR) to be higher. This section is followed by Chapter 3, which describes the methodology used in the analysis.

Chapter 3

Methodology

This chapter describes the methodology used to estimate the initial safe withdrawal rate for a Czech retail investor with a multi-currency portfolio and to compare the findings to the originally found 4% Rule by Bengen (1994). First, it describes the assumptions under which the thesis computes the rate and also the settings, i.e. the safe withdrawal rates investigated, the portfolio structures investigated and, among others, the portfolio's initial absolute value used in the computations. The chapter follows with a section about the steps and methods taken during modelling. It describes how data is processed, the structure of Monte Carlo Simulation (MCS) and what software is used during modelling.

3.1 Settings and Assumptions

A building block of the analysis is an investment portfolio. It is constructed of 50% stocks and 50% bonds, and this allocation is fixed. It could be possible to investigate different shares, but this approach is chosen to compare the results with the original study. Nevertheless, the structure of each of the assets varies among the portfolios considered. Both stocks and bonds vary in their share invested in CZK, EUR, and USD, i.e., in the Czech, Eurozone, and US markets. The exact shares are displayed in Table 3.1.

Table 3.1: Investigated currency splits for a Czech investor's portfolio

Scenario	**CZK**	**USD**	**EUR**
1. Full Home	100%	0%	0%
2. Moderate Diversification	50%	25%	25%
3. Equal Split	33%	33%	33%
4. Full Foreign	0%	50%	50%

These four splits are believed to be representative of international diversification within the set boundaries because they cover four major ways the basket can be cut. It represents a scenario where the investor is invested in the Czech market only—in this case, the exchange rate does not play a role, however, the domestic market attributes to some extent differ from the attributes of foreign ones, hence it is worth considering this case as well. An opposite extreme considered is a *Full Foreign* scenario where the retiree performs on outside markets only. For the simulation, this case is quite similar to the remaining two cases that consider different shares of investments among all three markets. Exchange rate risk is present, and inflation has the same properties since we still assume the rentier is located in the Czech Republic.

The optimal value of the portfolio varies based on individual needs and approach to FIRE. For the purpose of calculations, it is assumed that a retiree aims for monthly rent to be at least as much as the average salary. In 2023, the average gross monthly wage was CZK43,120 (Czech Statistical Office, 2024a). Assuming further that the calculated initial withdrawal rate is close to Bengen's rate, the optimal value of the investment portfolio may be $(43,120 \cdot 12)/0.04 = 12,936,000$ Czech korunas for some of the retirees. For better clarity, the initial value of the portfolio used in the computations is CZK15,000,000. The taxation of wages and income from investment portfolios may be different. Nevertheless, this is only used for the computations as an illustrative example. Since the withdrawn amount is a percentage, the absolute value of the portfolio does not affect the survival rates. If one's expenditures differ from the average wage, it is not difficult to calculate the necessary starting value of their investment portfolio.

The investigated initial withdrawal rates are summarised in Table 3.2. The table also describes a necessary portfolio worth at the beginning of the retirement period, rounded to the nearest thousand, to be able to withdraw the average salary described in the preceding paragraph. In other words, it shows how much the investor has to accumulate for different rates and therefore risks of premature depletion while keeping the living standard constant. Note that the higher the withdrawal rate, the higher the risk of depletion.

In parallel with empirical market performances, returns differ in the MCS throughout the payout period for all assets. Therefore, the modelling includes yearly rebalancing. If after a year, the stocks performed better compared to bonds in a given year and the former takes up 60% of the total amount invested in a given market, 10% of the stocks is sold and used to buy bonds to keep the

Table 3.2: Withdrawal rates and portfolio worth in CZK

Withdrawal Rate	**Portfolio Worth**
2%	25,872,000
2.5%	20,698,000
3%	17,248,000
3.5%	14,784,000
4%	12,936,000
4.5%	11,499,000
5%	10,348,000

ratio constant. Rebalancing is considered only on the instrument level, not on the market level. If the US market performs better than the Czech one, the proportion invested in each of the markets is not adjusted. Rebalancing within instruments keeps the portfolio's risk constant over the years. Rebalancing among markets, on the other hand, is not considered.

Safe withdrawal rate can be an ambiguous term. It could be understood as a percentage that gives the portfolio a 100% chance of surviving the whole payout period, or, in this case, 30 years. The lowest such percentage is called SAFEMAX (Bengen, 1994). Other studies, including the Trinity study (also, e.g., Pfau, 2010; Clare *et al.*, 2017), consider an initial withdrawal rate to be safe if it results in the portfolio outliving the payout period in 95% of the cases. This thesis inclines to the less strict definition and considers the latter case when interpreting the results.

The withdrawal strategy has been mentioned before. However, it is stated in this section as well for better clarity. The retiree withdraws a certain percentage of the total value of the portfolio in the first year. Then, every other upcoming year, they withdraw the same amount, not a percentage, only adjusted for inflation. The withdrawals happen once a year and last optimally the length of the payout period or until the portfolio is depleted, whichever happens first.

As far as inflation goes, it is treated in two ways. The first and main way is the standard inflation obtained from CPI. The second method accounts for the fact that the Czechs typically own a house (76% of the Czech population are homeowners compared to 65.7% of the US population, as of 2023; Statista, 2025) so their consumer basket has slightly different properties, it is not affected by the movements in rent prices. Hence, the inflation used in the calculations also includes a case where it does not contain housing. Furthermore, since housing is not a liquid asset and its loss would imply a necessary increase in spending, it is not represented in the portfolio either, nor is the opportunity cost of not

renting the property. It is rather assumed that the retiree benefits from the lack of rent burden and uses the additional funds either for the maintenance of the housing or to cover their spending elsewhere.

The modelling carries additional assumptions. One of them is the existence of all instruments throughout the whole retirement. The strength of this particular assumption varies across the variables. While it is reasonable to assume USD and US stocks and bonds indices will exist in the next 30 years, the chance of CZK as a currency lasting this long is questionable since the adoption of EUR is one of the requirements of the Czech Republic's membership in the European Union (EU) (Czech National Bank, 2025). Another assumption is that all withdrawals do not include tax deductions or any other fees. That is more important for the interpretation of results than for the calculations.

3.2 Monte Carlo Simulation

A method simulating numerous scenarios that can happen in the markets is necessary to get robust results. Such a method used in this thesis is called the Monte Carlo Simulation. Even though Bengen's original paper worked with historical backtesting, more recent literature (e.g. Guyton, 2004; Tamimi *et al.*, 2024) prefers MCS. In this case, the latter approach is also more convenient due to the limited availability of some data.

The steps of the simulation are as follows. First, relevant data is gathered. The thesis works with logarithmic returns for all but inflation data. Inflation is computed using the arithmetic return of the CPI to stay consistent with the prevailing approach. More information about individual variables is in Chapter 4. Then, the data is investigated for means and variances, and a covariance matrix is constructed of them all but inflation. Inflation is compiled from a consumer basket and fitted with log–normal distribution since the possibility of negative inflation, or deflation, is practically impossible.

The logarithmic returns are obtained in the following way:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

where

- r_t is the logarithmic return r in the year t,
- P_t is the price of the given variable in the year t.

Naturally, the first observation of every sample is lost due to the calculation. The reported sample periods in Chapter 4 do not include the lost year, i.e. they include only those years for which the logarithmic returns are available. It is worth mentioning that the thesis always works with logarithmic returns, except for inflation, which is referred to by its name. The terms logarithmic return, log return, and return are used interchangeably throughout the text, always meaning the first of these.

During the actual simulation, returns for every variable are generated from the standard normal distribution, multiplied by a Cholesky Decomposition matrix, and added to the means of the corresponding variables. Note that the Cholesky Decomposition matrix is always constructed only of the used assets and exchange rate, when applicable, and ensures that the drawn returns follow a similar correlation among each other as the empirical data. In other words, during simulation in *Full Home* scenario, the correlations of Czech stocks or Czech bonds with, e.g., US stocks are not taken into consideration. Showing returns generation in an equation:

$$\mathbf{r} = \boldsymbol{\mu} + \boldsymbol{L}\mathbf{z}$$

where:

- $\mathbf{r}$ is a vector of returns of the assets,
- $\boldsymbol{\mu}$ is a vector of means of the assets,
- $\boldsymbol{L}$ is a Cholesky Decomposition matrix,
- $\mathbf{z}$ is a vector of i.i.d. samples from standard normal distribution.

Cholesky Decomposition matrix is constructed such that the following applies:

$$\boldsymbol{\Sigma} = \boldsymbol{L}\boldsymbol{L}^T$$

where:

- $\boldsymbol{\Sigma}$ is a covariance matrix,
- $\boldsymbol{L}$ is the Cholesky Decomposition matrix.

Inflation is excluded and modelled independently due to the difference in the used distributions. Individual years are modelled independently as well,

however, the drawn returns at time $t-1$ affect the portfolio's value and structure at time t. After every modelling run for a given year, the inflation–adjusted amount is withdrawn, and then the portfolio is rebalanced within instruments. The amount is withdrawn proportionally from all instruments and markets, considering only those with positive balances.

To obtain statistically significant results, the simulation in every setting must be run an appropriate number of times. To obtain a 95% confidence over the results, the thesis employs a formula described by Glasserman (2003). The formula is following:

$$n = \left(\frac{z \cdot s_{total}}{E} \right)^2$$

where:

- n is the number of runs,
- z is the z-score corresponding to the desired confidence level (in this case: $z = 1.96$ for 95% confidence level),
- s_{total} is the estimated standard deviation across the simulations,
- E is the error rate.

The standard deviation s_{total} is calculated as the standard deviation of the portfolio value in the final year, where if the portfolio was depleted prematurely, the final value is set to 0, assuming one cannot get into debt on their investment portfolio. The resulting n's are obtained from 500 initial MCS runs that do not count towards the final number of runs in the specific setting. The error rate E is set to CZK300,000, corresponding to half of the withdrawn amount, or 6–month salary, in the initial year. Then, even if the error occurred to the full extent, it would not add a whole year to the portfolio's longevity. It could still increase the longevity in a case where the true funds correspond to only, for example, 8 months and the error would cause it to last a full 12 months. Analogically, it could also decrease the time the portfolio lasts. The impact is assumed not to be dramatic, and hence, such E is considered reasonable.

For each of the portfolio's market splits, a unique minimum number of simulations is calculated. However, it is calculated with 4% initial withdrawal rate in every case and rounded up to the nearest ten thousand. The number of simulations n is summarised in Table 3.3. Higher n then symbolises higher

variance in final portfolio values. Note that every simulation (i.e. scenario–withdrawal rate combination) is initiated with the same seed to mitigate the differences emerging from randomness and provide a more accurate comparison.

Table 3.3: Investigated scenarios for portfolio simulations

Scenario	**Number of runs (n)**
1. Full Home	50,000
2. Moderate Diversification	50,000
3. Equal Split	60,000
4. Full Foreign	90,000

Finally, the results are interpreted. It is stated what implications the results have for a Czech retail investor with a similar setting, and the findings are also compared to the original ones estimated by Bengen (1994). The thesis shows to what extent the estimations differ and in what ways they are similar.

The analysis is done using programming language Python 3.11 and, among others, packages NumPy for modelling and Matplotlib for visualizations. Full list of packages and their versions used as well as the actual codes used during data processing and simulations are available upon request.

Chapter 4

Data Description

This thesis conducts a heavily data–dependent analysis. The periods for which the data is considered, the treatment of outliers, and the instruments used to represent given variables. All these influence the portfolio performance throughout the MCS, and because of that, it is critical to be direct with the approach used, including the limitations and the benefits it has.

All relevant variables are described in this section individually. Starting with inflation, through exchange rates and stocks and ending with bonds, every metric has reported descriptive statistics (mean and variance) and a histogram of its distribution. The histograms serve the purpose of showing the limitations or benefits of using normal distribution to model the returns. The closer the given histogram resembles a bell curve, the less of a limitation the normal distribution is. Moreover, the description always includes the period used and, if relevant, comments on outliers. Note that when the outliers are not mentioned, none were identified. The chapter ends with a description of the relationships between the variables, focusing on correlations.

4.1 Inflation

Inflation is present in generally any economy and had it not been included in the simulation, the market returns would be greatly overestimated. The data is obtained from Czech Statistical Office (2025) and treated according to the description below.

Throughout the years 1996–2017, inflation is measured using the CPI index calculated using the Classification of Individual Consumption by Purpose. For the rest of the observed period (2018–2024), the European Classification

of Individual Consumption According to Purpose was used (Czech Statistical Office, 2025). The modelling is then based on inflation measured between 1996 and 2024. The average and variance are reported in Equation 4.2.

$$\mu_\pi = 0.038, \sigma_\pi^2 = 0.0356 \tag{4.1}$$

It is assumed that housing prices rise more quickly than the rest of the consumer basket's attributes. This is empirically supported because, in the period of 1997–2024, average housing inflation was 4.66%, while average total inflation was just 3.62% (Czech Statistical Office, 2025). Combining this with the fact that Czechs often own their property, this thesis also investigates the cases when personal inflation is systematically underestimated by housing. According to Czech Statistical Office (2024b), housing constructs 10.34% of the total consumer basket, and its inflation was 1.12% in 2024. It contributed to the total inflation percentage with $\alpha = 0.1158$ percentage points at the time. The underestimated statistics are then reported in Equation 4.2.

$$\hat{\mu}_\pi = 0.0368, \hat{\sigma}_\pi^2 = 0.0356 \tag{4.2}$$

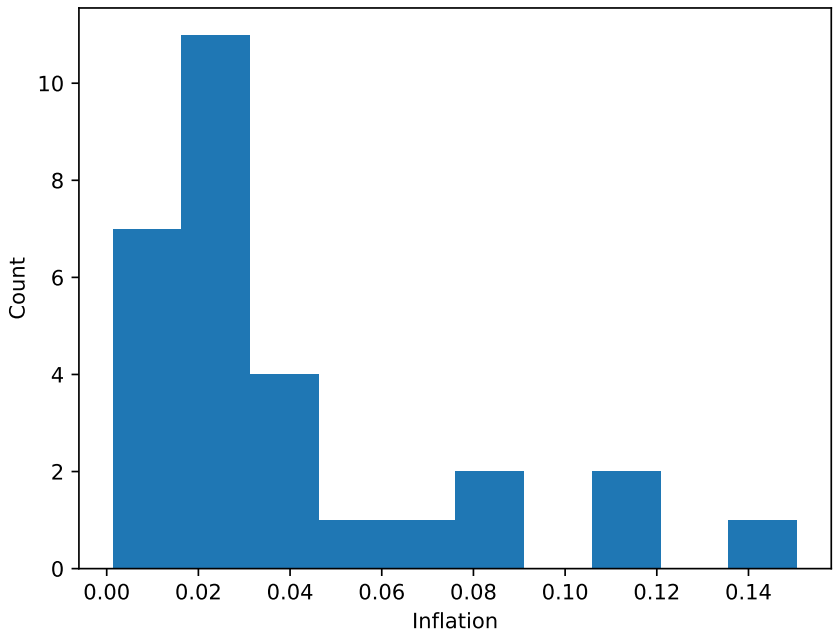


Figure 4.1: Histogram of annual inflation

Since the possibility of negative inflation, i.e. deflation, is virtually zero, the modelling assumes it follows a log-normal distribution. This is further supported by a histogram of historical yearly inflation as shown in Figure 4.1. Note

that the histogram shows not underestimated data. Moreover, the log-normal distribution is bounded in MCS from above to not generate values greater than 0.5, i.e. 50%. The probability of such an event happening is 0.22% given the distribution, which may not seem worth mentioning. However, it would occur 2 times for every 1,000 runs, and that is considered substantial. The thesis assumes currency devaluation cannot reach such extremes as a central bank would take actions to slow the process down. While events above, for instance, 30% are also unlikely, they are considered to be possible under special circumstances.

By doing a scan for outliers using the Z–Score method, it was found that the year 2022 with inflation 15.06% has $z_{\pi,2022} = 3.17$. Nevertheless, this inflation is considered reasonably possible. It is kept in the sample, and the reported statistics are affected by the value.

4.2 Exchange Rates of Czech Koruna to Other Currencies

The presence of exchange rates is a vital part of the analysis. Since the three markets considered have all different currencies, the returns and, therefore, also the total amount invested in foreign markets depend on the current exchange rate between domestic (CZK) and external (EUR, USD) currencies. This section describes where the data on exchange rates were obtained from, how they were processed, its distribution and statistical properties. In both cases, CZK is considered as the base currency, i.e. the data shows the logarithmic change of *how much EUR/USD costs in terms of CZK*.

Both exchange rates were obtained from Refinitiv Eikon (2025) on a yearly basis on the first work day of the year.

4.2.1 Czech Koruna to Euro

The first described spot exchange rate is the euro to the koruna. Because the euro launched after the koruna, on 1 January 1999, only the period 1999–2025 is considered. In consequence of the dates being obtained on the first workday of every year, the year 1999 is represented by the exchange rate obtained on 3 January 2000, and therefore, the base year is not lost during the transformation

to returns. The final year is 2024, and the total sample then consists of 26 data points.

The computed mean of returns and variance are reported in Equation 4.3. Even though the statistics are indexed with EUR tag, the numbers correspond to the values of the log rate of returns of EUR/CZK. The average exchange rate return is negative, presumably because of the convergence of EUR and CZK since the launch. Interestingly, σ^2_{EUR} is very low.

$$\mu_{EUR} = -0.0139, \sigma^2_{EUR} = 0.0014 \tag{4.3}$$

The histogram of this exchange rate's distribution is described in Figure 4.2. It appears the data fit the normal distribution constructed upon them quite well.

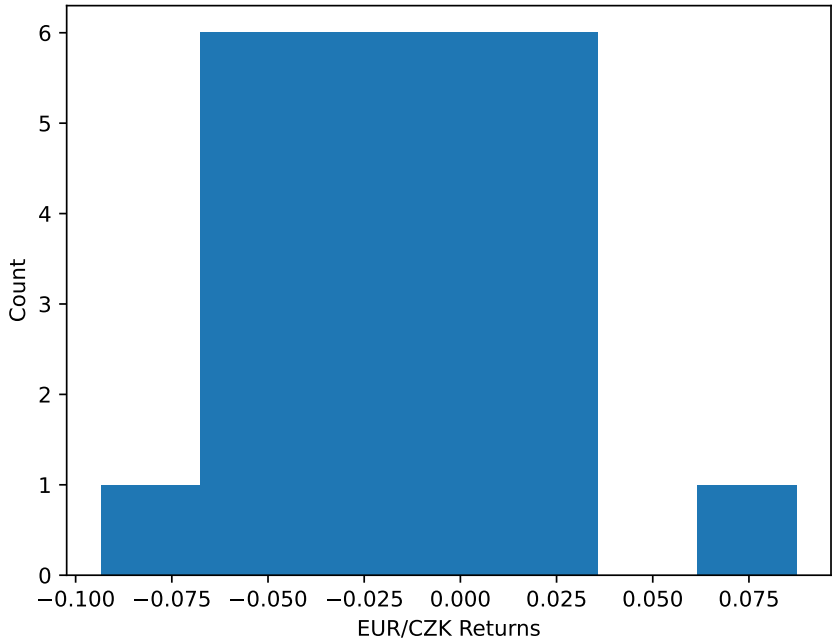


Figure 4.2: Histogram of rates of returns of EUR/CZK exchange rate

4.2.2 Czech Koruna to Dollar

The other exchange is US dollar to Czech koruna. Here, the data is limited by the existence of the Czech Republic, i.e. by 1 January 1993. Data is obtained in the period from 1993 to 2025. Due to the same reasons as in the case of the Czech koruna to the euro, the final sample consists of data from 1993 until 2024.

It is worth noting that the calculated mean is very close to 0, as seen in Equation 4.4. On the other hand, its variance is almost eight times higher compared to EUR/CZK exchange rate. The histogram of the distribution of the data is displayed in Figure 4.3.

$$\mu_{USD} = -0.007, \sigma^2_{USD} = 0.0105 \tag{4.4}$$

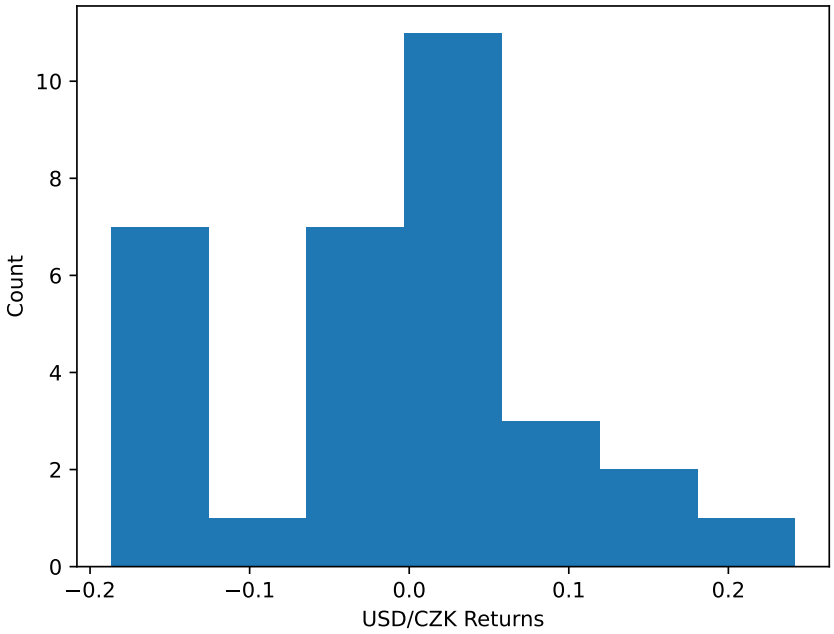


Figure 4.3: Histogram of rates of returns of USD/CZK exchange rate

4.3 Stocks

This section describes data on stocks used in the modelling. All stock data were obtained from Refinitiv Eikon (2025).

Rates of return of given stock indices are collected with yearly frequency, always as of 31 December in the given year, and the cut–off point of the observed periods is the end of 2024. The start point differs among markets because the indices vary in their creation date.

For Czech and American stocks, total log return versions of indices are used, and gross log return is used for European, or more specifically, Eurozone, stocks. Both total and gross returns include the price changes as well as reinvested dividends, but gross returns do not deduct taxes on the dividends. The reason for this inconsistency is the limitation of available data. Nevertheless,

these forms of indices better represent the real performance of given markets compared to price return indices, since market performance is also reflected in dividends.

4.3.1 Czech Stocks

To represent stocks of the Czech market, the PX–TR index (ticker: PXTR) by the Prague Stock Exchange is used. The index started in March 2006, and hence, the data available spans from 2007 until 2024. From the available 18 years long period, the following parameters are obtained:

$$\mu_{PXTR} = 0.1016, \sigma^2_{PXTR} = 0.0257 \tag{4.5}$$

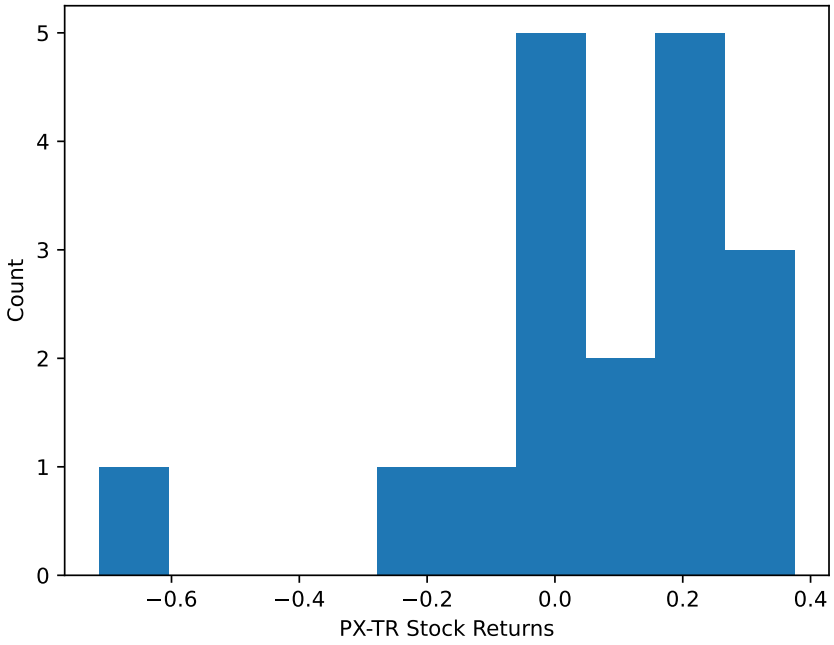


Figure 4.4: Histogram of rates of total returns of PX–TR index

Even though the history is fairly limited, there are years with significant (under) performance of the index. The steepest plummet happened in 2008 when the index decreased by -71.28%. Nevertheless, this observation is an outlier in the sample and is excluded from the reported statistics but included in the histogram. The worst–performing non–outlier year was 2011, with a decrease of -23.94%. The largest rise happened in 2021, with an increase of 37.45%. The histogram of the distribution of returns is displayed in Figure 4.4.

Given that the index is fitted with a normal distribution following the empirical statistics, the probability of a shift at least as dramatic as the observed maximum (minimum) is 4.43% (1.67%). This can be considered a limitation of the approach used. However, the number of simulations is high enough that these events are likely to occur in some of the runs.

4.3.2 European Stocks

The European market stocks are represented by the gross logarithmic return of the EURO STOXX 50 index (ticker: SX5GT) created by STOXX company. The index tracks supersector leaders in the Eurozone and hence is a great representation of the Eurozone market (STOXX, 2025). The data used spans from the year 2001 to 2024. The empirically obtained descriptive statistics mean and variance are displayed in Equation 4.6, and a histogram of the distribution is displayed in Figure 4.5.

$$\mu_{SX5GT} = 0.0835, \sigma^2_{SX5GT} = 0.0174 \tag{4.6}$$

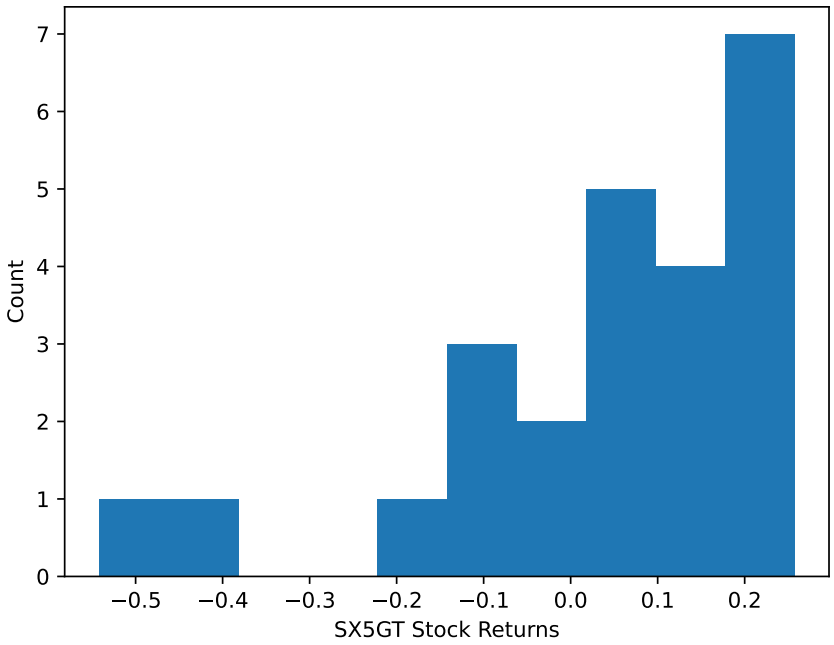


Figure 4.5: Histogram of rates of gross returns of EURO STOXX 50 index

Similarly to the PX–TR index, there are some extreme values in the sample. The years 2002 and 2008 carried returns of -44.2% and -54.16%, respectively,

and both are removed outliers. The largest decrease kept in the sample is in 2001, with a decrease of -20.82%. On the contrary, an increase of 25.7% happened in 2019 is the best in the sample. While the probability of year–to–year increase being at least as large as the empirical maximum is 9.42% with $\mathcal{N}(\mu_{SX5GT}, \sigma^2_{SX5GT})$ distribution, the probability of decrease at least as serious as the empirical minimum is only 1.35%. Hence, the latter event occurs in fewer runs.

4.3.3 American Stocks

American stocks are represented by total logarithmic returns of the S&P 500 index (ticker: SPXTR) by S&P Dow Jones Indices. The index covers five hundred leading companies of the US markets, effectively covering around 80% of local market capitalisation. Therefore, it serves as a valid representation of the market (S&P Global, 2025). The available data are obtained from 1988 to 2024, and the descriptive statistics are displayed in Figure 4.6.

$$\mu_{SPXTR} = 0.1218, \sigma^2_{SPXTR} = 0.0205 \tag{4.7}$$

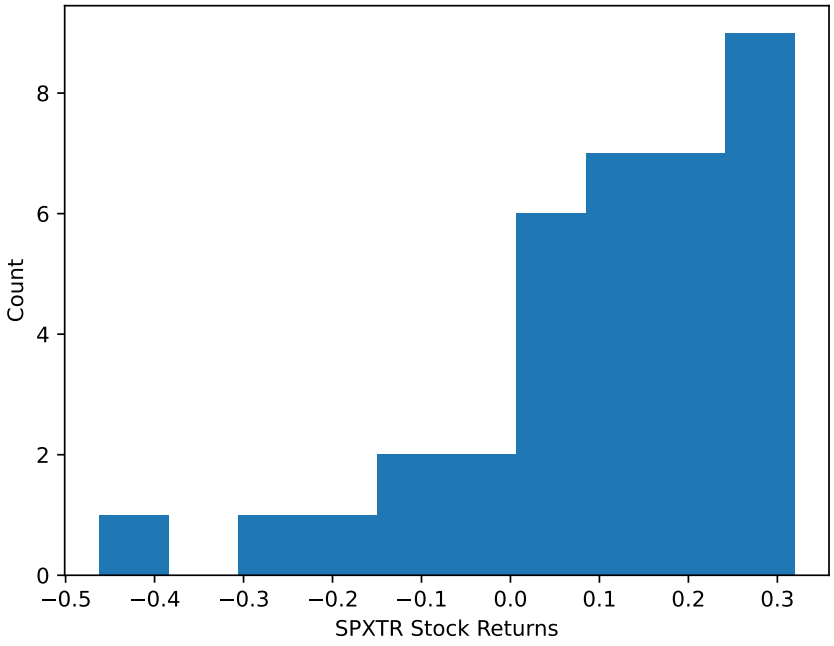


Figure 4.6: Histogram of rates of total returns of S&P 500 index

Vis-à-vis the returns, the worst year was again the year 2008, when the index decreased by -46.2%. Even in this case, it is an outlier and is dropped

from the sample. The worst performing year kept in the sample is 2002, with a decrease of -24.98%. On the other hand, the best year was 1995, when the index increased by 31.9%. It is evident from the histogram that the index had positive returns in the majority of the years, and in consequence, the probability of an event at least as bad as, to this day, the deepest plummet is only 0.47%. The probability of a yearly return being no worse than the return in 1995 is 8.42%.

4.4 Bonds

Bonds are just as important for the analysis as stocks because the portfolio is constructed of an equal combination of the two. Compared to its counterpart, bonds are generally *safer* investment as their variance tends to be lower. On the other hand, their returns are also usually below those of stocks.

All three markets are represented using government bond total returns indices created by FTSE International Limited, collected from Refinitiv Eikon (2025). The indices track local currency, fixed–rate, nominal government bond markets (FTSE RUSSELL, 2025). More information about individual characteristics, including Reuters Instrument Code (RIC) for better identification, is mentioned in the individual sections below.

Data are obtained with a yearly frequency. The endpoint is always 2024. However, the length of the sample period varies among countries based on the availability of data. Similarly to stocks, the total return of bonds is used to measure the total performance of the bonds, reflected in both bond price and paid–out coupons.

4.4.1 Czech Bonds

The Czech Republic government bond return index (RIC: FTCZ_TSYLCLT) represents the bonds in the retiree's local market. Only non–zero–coupon bonds with issues larger than CZK15 billion are considered in the index. The final sample period includes data from 2012 until 2024, and the descriptive statistics are displayed in Equation 4.8.

$$\mu_{CZBond} = 0.0176, \sigma^2_{CZBond} = 0.0047 \tag{4.8}$$

A histogram of data on Czech bond log returns is displayed in Figure 4.7. The lowest return happened in 2021 with. Conversely, the highest return hap-

pened in 2012 with $r_{CZBond,2012} = 12.44\%$. The probability of such returns happening, given the normal distribution with empirically obtained parameters, is 3.85% and 5.98%, respectively.

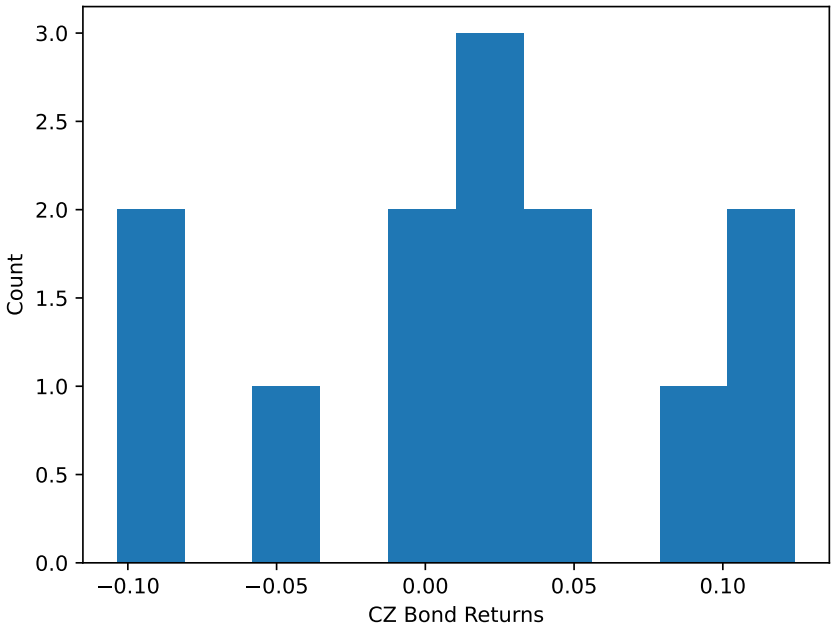


Figure 4.7: Histogram of Czech government bond index returns

4.4.2 European Bonds

In the case of the Eurozone, the representation of one government bond is not so straightforward. Since the Eurozone lacks one common bond, the market has to be represented in a different way by a particular country. Were the decision to be made on the government's credit quality or liquidity, France would probably be a better choice. However, current literature often picks Germany. The country's government is considered reliable and stable, and its debt is relatively sustainable. Not only for these reasons is the German 10–year government bond currently a widely accepted financial benchmark of the Eurozone (Preunkert, 2023). With the intention to stay consistent with the literature, this thesis picks Germany as well, even though the selected index (RIC: FTDE_TSYLCLT) also contains bonds of other than 10–year maturity. Note that it is referred to the bonds as European or EU bonds in the descriptive statistics' indices and histogram as well as in the later text. This is because, in this case, the representation is more important than the background data, and to avoid confusion.

$$\mu_{EUBond} = 0.0427, \sigma^2_{EUBond} = 0.0019 \tag{4.9}$$

The sample includes a specific list of specific categories of bond instruments issued with a size of at least EUR2.5 billion in the period of 1994 to 2024. (FTSE RUSSELL, 2025). The data include an outlier, the year 2022, with the return of $r_{EUBond,2012} = -19.67\%$. This observation was removed from the data and is included only in Figure 4.8. The descriptive statistics in Equation 4.9 are not influenced by the year.

Apart from the outlier year, the lowest return happened in the year 2013 with a decrease of -2.73%, and the highest one happened 26 years earlier with an increase of 15.08%. The returns follow a bell curve quite well when the leftmost observation is not considered. Even so, the probabilities of returns at least as dramatic as the non–outlier extremes are not substantially higher compared to other variables (5.42% for minimum, 0.66% for maximum).

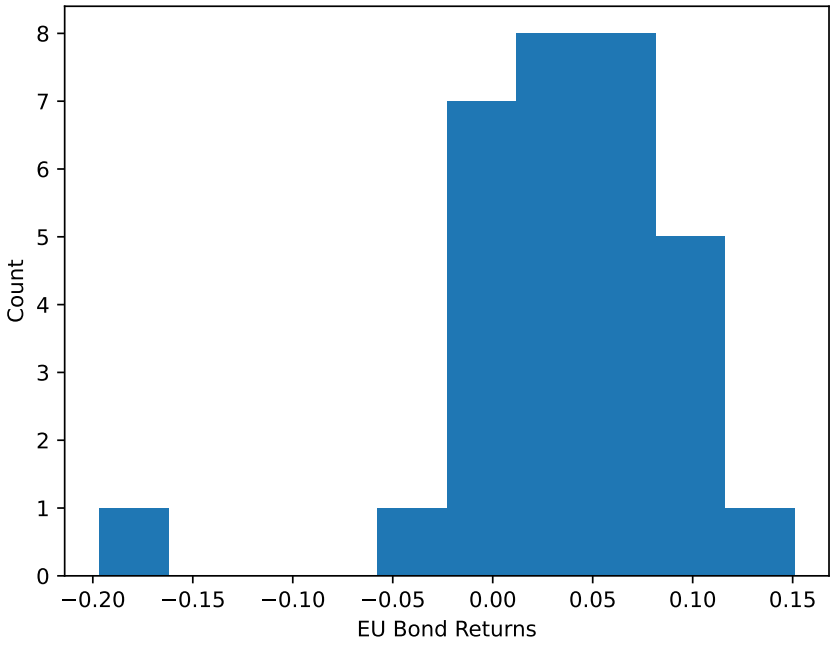


Figure 4.8: Histogram of European government bond index returns

4.4.3 American Bonds

American bond market is also represented by local government bonds (RIC: FTUS_TSYLCLT). Nevertheless, the index for US consists of bonds issued with a minimal size of USD5 billion public amount outstanding (excluding Fed-

eral Reserve holdings). The index also includes callable bonds. Similarly to European bonds, the sample period lasts from 1994 to 2024. The descriptive statistics are displayed in Equation 4.10, and a histogram of the distribution of US bond logarithmic returns is displayed in Figure 4.9.

$$\mu_{USBond} = 0.0384, \sigma^2_{USBond} = 0.0036 \tag{4.10}$$

Similarly to the European bond index and US stocks, the best–performing year for the asset was 1995, when the returns reached 16.81%. Building on the similarity with the European bonds, the US government bond index had its worst year in 2022 (decrease by −13.48%). However, unlike EU bonds, it is not an outlier in the sample. $\mathcal{N}(\mu_{USBond}, \sigma^2_{USBond})$ distribution generates at least as good (poor) returns in 1.53% (0.19%) of cases.

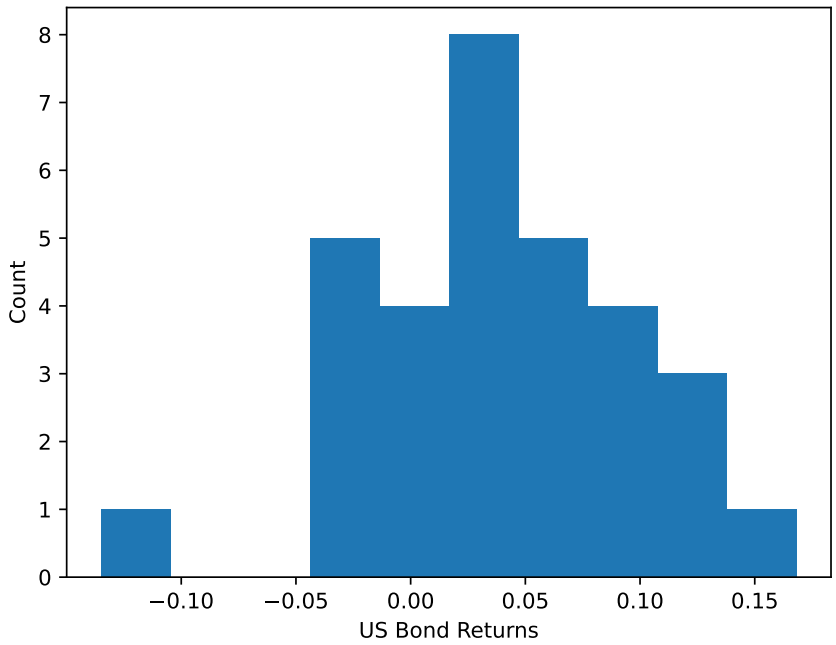


Figure 4.9: Histogram of American government bond index returns

4.5 Relationships Among Variables

To ensure the simulation reflects factual relationships between the variables, the Monte Carlo Simulation employs Cholesky Decomposition obtained from a correlation matrix. The correlation matrix is displayed in Figure 4.10.

It can be seen that the strongest relationships appear between the same types of assets. Interestingly, the contest for the strongest Pearson's correlation

coefficient is tied between the European bond index with the Czech bond index and the American bond index ($\rho_{EUBond,CZBond} = \rho_{EUBond,USBond} = 0.81$).

It is worth mentioning that the EU bond index negatively correlates with all three stock indices considered. When compared to the Czech bond index that shares a positive relationship with two of the stock indices, the EU bond index has a stronger correlation in two of the three cases, even though it is negative.

Apart from that, Pearson's correlation coefficient across asset types is not very strong. For example, the EURO STOXX 50 index with EUR/CZK exchange rate and the EU bond index with USD/CZK exchange rate show almost no connection at all the corresponding ρ being very close to zero.

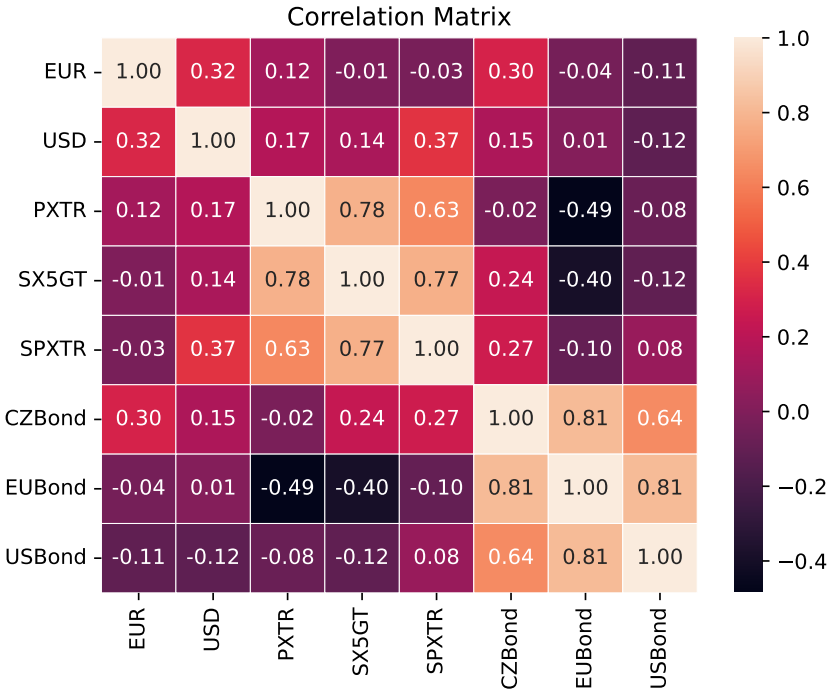


Figure 4.10: Correlation matrix of relevant variables

Note that the variables *PXTR* and *CZBond* belong to the Czech market, *SX5GT* and *EUBond* represent the Eurozone, *SPXTR* and *USBond* correspond to the US, and EUR and USD show the exchange rates between the Czech koruna and the euro, and the dollar, respectively.

Chapter 5

Results and Discussion

This chapter describes the results of the simulation defined above. It answers whether the 4% Rule holds for a Czech retail investor. Portfolio performances under the relevant settings are reported, and the most important outcomes are highlighted and compared to those described in Chapter 2, to the results originally found by Bengen (1994), and to the more optimistic or pessimistic ones reported by others. It concludes the analysis, hints at other insights that are implied by the results and discusses how the Czech investor may obtain higher withdrawals than those reported.

5.1 Main Findings

Table 5.1: Success rates of initial withdrawal rates under different scenarios and total inflation

	Initial Withdrawal Rate						
Scenario	**2%**	**2.50%**	**3%**	**3.50%**	**4%**	**4.50%**	**5%**
1. Full Home	**99.61%**	**97.69%**	**92.64%**	**83.39%**	**70.04%**	**54.66%**	**40.10%**
2. Moderate Diversification	**99.81%**	**98.66%**	**94.83%**	**86.78%**	**74.58%**	**59.30%**	**43.64%**
3. Equal Split	**99.74%**	**98.49%**	**94.55%**	**86.52%**	**74.35%**	**59.32%**	**44.41%**
4. Full Foreign	**99.48%**	**97.55%**	**92.81%**	**84.16%**	**71.99%**	**58.27%**	**44.63%**

The survival rates with total inflation are summarised in Table 5.1. It can be immediately observed that *the safe withdrawal rate*, which does not deplete the portfolio prematurely in 95% of the cases, is somewhere between 2.5% and 3% for all scenarios considered. Since the estimation is not based on a theoretical model of which a derivation could be taken, the exact safe rate could be obtained using a trial–and–error method only. However, that would

Table 5.2: Portfolio's terminal value statistics per CZK1,000 invested (real value; total inflation)

		Initial Withdrawal Rate						
Scenario	**Statistic**	**2%**	**2.50%**	**3%**	**3.50%**	**4%**	**4.50%**	**5%**
1. Full Home	Mean	**1,473**	**1,232**	**995**	**769**	**574**	**407**	**276**
	Max	**20,553**	**14,179**	**13,156**	**12,641**	**10,911**	**13,369**	**9,769**
	Min	**0**	**0**	**0**	**0**	**0**	**0**	**0**
	Median	**1,193**	**962**	**738**	**512**	**283**	**66**	**0**
2. Moderate Diversification	Mean	**1,567**	**1,319**	**1,074**	**837**	**627**	**441**	**296**
	Max	**13,755**	**17,646**	**14,807**	**14,382**	**11,789**	**13,882**	**12,419**
	Min	**0**	**0**	**0**	**0**	**0**	**0**	**0**
	Median	**1,302**	**1,063**	**831**	**597**	**365**	**130**	**0**
3. Equal Split	Mean	**1,594**	**1,343**	**1,097**	**862**	**650**	**464**	**318**
	Max	**15,835**	**18,263**	**17,941**	**13,675**	**11,149**	**11,174**	**13,665**
	Min	**0**	**0**	**0**	**0**	**0**	**0**	**0**
	Median	**1,314**	**1,077**	**838**	**607**	**375**	**141**	**0**
4. Full Foreign	Mean	**1,670**	**1,413**	**1,166**	**927**	**710**	**526**	**375**
	Max	**23,491**	**28,883**	**33,335**	**17,365**	**29,947**	**14,814**	**15,653**
	Min	**0**	**0**	**0**	**0**	**0**	**0**	**0**
	Median	**1,313**	**1,072**	**835**	**597**	**369**	**133**	**0**

result in an SWR specified to so many decimal places that it would be seed sensitive, even with the large number of runs specified in Table 3.3. Instead, it is concluded that the *safe* rate, the FIRE Rule for a Czech retail investor, is the more conservative bound, 2.5%.

Comparing the results with other studies, they are not as buoyant as those estimated by Bengen (1994) or Cooley *et al.* (1998), nor as gloomy as those calculated by, e.g., Pfau (2014) for some countries. Nevertheless, the 4% Rule is not validated from the perspective of a Czech retail investor because portfolios outlast the 30–year–long payout period in only around 70–75% of the cases across all scenarios with such a rate. Additionally, when withdrawals as high as 5% are considered, the portfolio constructed of 50% stocks and 50% bonds does not survive at least 30 years in around every second simulation considered in all market split settings.

An interesting property of the results is that the decrease in survival rate with an increase in withdrawals has a differing pace across both scenarios and withdrawal rates. While the highest survival rate for a 2% rate is in the *Moderate Diversification* case, *Full Foreign* portfolio performs the best with a 5% rate, compared to other splits, suggesting the latter scenario's performance decreases more slowly compared to the former one. *Moderate Diversification* also contains the largest and the smallest changes in performance with an increase in withdrawals. The rise from 2% to 2.5% worsened the performance by only

1.15 percentage points, but a further increase from 4.5% to 5% caused a decrease by 15.66 p.p. The largest and the smallest changes happened with the same changes in rates in all scenarios, just of lesser magnitude.

The results are improved by international diversification. The benefits outweigh the negative exchange rate returns of EUR/CZK and USD/CZK, and in consequence, the portfolios active in more than one market almost always perform better than the *Full Home* case involved in only one market. The only exception is with a 2% and 2.5% rates, where the domestic market outperforms an equal split of foreign markets by 0.13 and 0.14 percentage points, respectively, ranking third among the scenarios.

To provide a better understanding of the portfolio performance under the considered framework, Table 5.2 displays statistics per every CZK1,000 invested in given settings. Moreover, the average performances across runs, scenarios and withdrawal rates are displayed in Section A.1. The minimal terminal value is zero for every case because none had a survival rate of 100%. Similarly, the median value is CZK0 for every 5% withdrawal rate because the portfolios are depleted prematurely in the majority of the cases in this setting.

The best-off portfolios are always identified under the fourth scenario. However, the median value in the *Full Foreign* is never the highest. This is caused by the comparably high variance already suggested by the number of runs n of the Monte Carlo Simulation in Table 3.3. On the other hand, the lowest median is connected to *Full Home*, no matter the withdrawal rate. By comparison of the two scenarios, it is clear that the higher returns of the portfolio are redeemed by higher variance, as is customary in financial practice.

When the median value is above the start value (i.e., above CZK1,000), as it is for 2.5% rate, it may be argued that the withdrawal rate is too low since the portfolio's worth increased during the payout period. However, the 3% rate is then too high such that the survival rate does not reach the requirements to be considered safe. This argues for the exact SWR to be between the rates.

5.2 Sensitivity to Inflation

The survival rates with underestimated inflation are reported in Table 5.3. While the results improve by only tenths of a percentage point, and for *Moderate Diversification* split and 2% withdrawal rate, the survival rate stayed the same, compared to the results in Table 5.1, the changes are serious enough to improve the safe withdrawal rate in two of the scenarios. When the Czech

Table 5.3: Success rates of initial withdrawal rates under different scenarios and underestimated inflation

	Initial Withdrawal Rate						
Scenario	**2%**	**2.50%**	**3%**	**3.50%**	**4%**	**4.50%**	**5%**
1. Full Home	**99.66%**	**97.96%**	**93.39%**	**84.47%**	**71.62%**	**56.48%**	**41.90%**
2. Moderate Diversification	**99.81%**	**98.85%**	**95.36%**	**87.88%**	**76.31%**	**61.21%**	**45.91%**
3. Equal Split	**99.79%**	**98.68%**	**95.04%**	**87.62%**	**76.05%**	**61.44%**	**46.37%**
4. Full Foreign	**99.53%**	**97.83%**	**93.25%**	**85.14%**	**73.53%**	**60.19%**	**46.19%**

Table 5.4: Portfolio's terminal value statistics per CZK1,000 invested (real value; underestimated inflation)

		Initial Withdrawal Rate						
Scenario	**Statistic**	**2%**	**2.50%**	**3%**	**3.50%**	**4%**	**4.50%**	**5%**
1. Full Home	Mean	**1,535**	**1,289**	**1,045**	**820**	**613**	**441**	**299**
	Max	**21,430**	**14,690**	**13,663**	**16,533**	**16,475**	**12 703**	**12,441**
	Min	**0**	**0**	**0**	**0**	**0**	**0**	**0**
	Median	**1,242**	**1,014**	**784**	**549**	**320**	**98**	**0**
2. Moderate Diversification	Mean	**1,633**	**1,379**	**1,129**	**888**	**669**	**475**	**321**
	Max	**22,011**	**19,098**	**11,682**	**14,227**	**10,757**	**15,789**	**13,223**
	Min	**0**	**0**	**0**	**0**	**0**	**0**	**0**
	Median	**1,358**	**1,117**	**876**	**640**	**405**	**166**	**0**
3. Equal Split	Mean	**1,661**	**1,403**	**1,155**	**913**	**692**	**499**	**344**
	Max	**16,449**	**16,629**	**14,937**	**13,369**	**11,037**	**12,151**	**11,391**
	Min	**0**	**0**	**0**	**0**	**0**	**0**	**0**
	Median	**1,374**	**1,130**	**888**	**647**	**410**	**175**	**0**
4. Full Foreign	Mean	**1,737**	**1,478**	**1,224**	**978**	**756**	**564**	**403**
	Max	**23 104**	**21,558**	**23,673**	**20,213**	**14 061**	**17,271**	**14,185**
	Min	**0**	**0**	**0**	**0**	**0**	**0**	**0**
	Median	**1,363**	**1,124**	**884**	**642**	**403**	**169**	**0**

investor is active in the Czech, the Eurozone, and the US markets and owns their own house, their *safe withdrawal rate* is 3%. When the retiree is active in the foreign markets only, or, vice versa, has all funds in the domestic market, the safe rate stays at 2.5% of the initial value.

The *Full Home* scenario now ranks the third best in three out of the seven withdrawal rates investigated. However, when it is not the worst, it is at most by 0.14 p.p. better than the worst scenario, which is *Full Foreign*. Moreover, the domestic market alone is still the worst in the majority of cases. The benefits of diversification are clearly visible.

Table 5.4 shows that the 3% withdrawal rate may be very close to the true parameter in the two scenarios. The median final value is below the initial value, suggesting the withdrawal rate is not too low, and the survival rate is above 95%, suggesting the withdrawal rate is not too high. On the other hand, the issue of the estimated safe withdrawal rate being too conservative due to the terminal value being unnecessarily large persists within the splits that do not consider at least one market.

The average performance of the portfolio in all considered settings with the inflation being underestimated by housing is displayed in Section A.2.

5.3 Implications for Practice

Due to *the safe withdrawal rate* being below the 4% rule of thumb, it may seem a Czech FIRE adherent would need higher savings than an adherent from the US to achieve the same retirement period length. However, the Czech Republic has a mandatory pension insurance system. During the saving period before leaving the labour market, the investor must be subscribed to the insurance and make payments (Czech Statistical Office, 2024c). Then they enjoy a monthly income once they reach the retirement age set by the government. This only applies, however, if the retiree spends at least 35 years paying the insurance (gov.cz, 2024).

The Czech retiree could then adopt a semi–dynamic approach to the retirement plan and combine the income from the investment portfolio and the income from the pension system. If the investor retires before the government–set age (the goal of the FIRE community), they would start with higher withdrawals from the investment portfolio, since the pension would be zero. Once they reach the retirement age, they rebalance the withdrawals from the investment portfolio to amount to the annual difference between the budget and the

pension. One way to calculate the real initial value of the available funds could be to sum the value of the investment portfolio and the total expected pension.

Chapter 6

Conclusion

The thesis intends to determine a safe withdrawal rate for a Czech retail investor active in up to three markets. The rate was originally estimated by Bengen (1994) to be 4% for a US retiree, thus creating the 4% Rule, or the FIRE Rule. The rule has not been previously estimated for a Czech investor, and the literature regarding the impact of international diversification on the withdrawal rate is not extensive. Hence, the thesis adds to the research in these two ways.

In total, four scenarios of diversification are considered: from activity on the Czech market only, i.e., no diversification, to no activity in the domestic market, i.e., equal split between the Eurozone and US markets. The third considered split has half of the funds invested in the retiree's own country and the rest equally divided between EU and US markets. The last scenario is an equal split of the three markets. However, the splits correspond to the initial year only. If, during the simulation, one market performs better than the other, the proportions are changed and not adjusted.

The considered portfolios are constructed of 50% stocks and 50% bonds in every scenario. The shares are annually rebalanced to stay consistent throughout the payout period. Because the Eurozone and the US operate in different currencies than the Czech koruna, the exchange rate is also included in the calculations. Furthermore, since it was found that the Czechs are likely to own a property (Statista, 2025), the thesis performs a sensitivity analysis by comparing the performance with total inflation to the performance with inflation underestimated by housing.

The estimation is approached through a Monte Carlo method, running a large number of simulations to obtain statistically significant results. The loga-

rithmic returns are generated from a normal distribution, employing Cholesky decomposition to capture the empirical correlation between the variables, except for inflation, which is modelled independently using a log–normal distribution.

The thesis finds the safe withdrawal rate to be at least 2.5% and at most 3% in all of the scenarios considered with a payout period of 30 years, concluding that the 4% Rule does not hold for a Czech retail investor. The highest safe withdrawals, 3%, are observed in portfolios active on all three markets when the retiree owns real estate. For the rest of the scenarios, the estimate SWR is 2.5%. The survival rate with a 4% withdrawal rate applied is around 70%–75%, depending on the market split. The highest terminal values of a portfolio are always associated with the portfolio that was not active on the Czech market. On the other hand, the highest median values were observed in the scenarios that consider all three markets. Despite the fact that the survival rates are similar in all market splits, the portfolio that is not internationally diversified performs the worst under the majority of withdrawal rates investigated. This implies that the results benefit from international diversification.

Future research could focus on the implementation of the pension system into FIRE rule calculations, as the literature regarding this matter is very limited. The retiree would then be able to enjoy higher withdrawals throughout the whole payout period. Another dimension to the calculation could be added by considering the possibility of the Czech Republic adopting the euro sometime during the payout period, as it is required to (Czech National Bank, 2025). This would mitigate the exchange rate risk between the Czech and the Eurozone markets and could again increase the withdrawal rates.

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Appendix A

Performance of Monte Carlo Simulations

The appendix shows average portfolio performance under different scenarios and withdrawal rates. It is sectioned into two parts. The first one shows results with the total inflation. The second part is focused on the simulations where inflation is underestimated. The portfolio worths are expressed in real rather than nominal terms. Note that the first withdrawal happens before the first returns, which explains the initial dip in all of the graphs.

A.1 Total Inflation

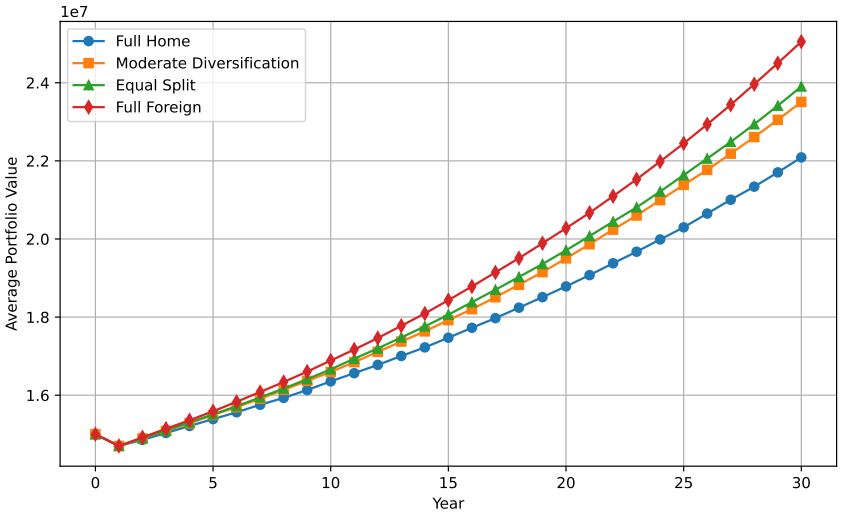


Figure A.1: Average portfolio value over time (Rate: 2%; total inflation)

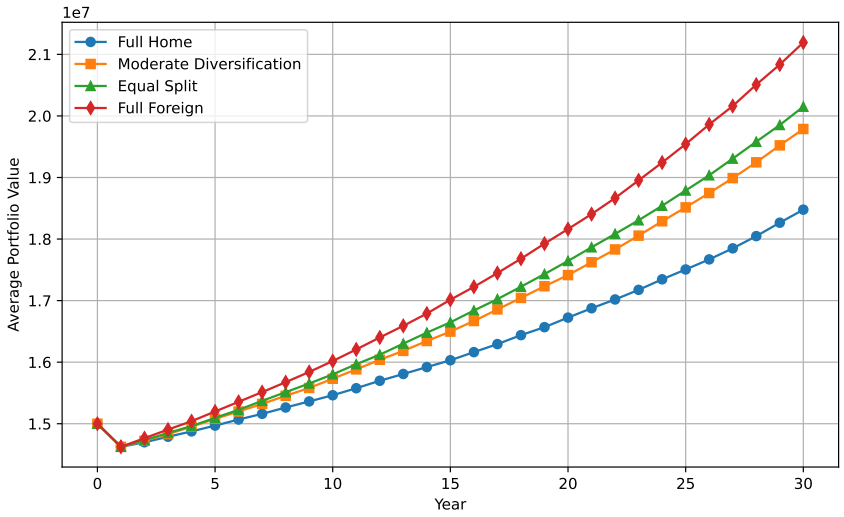


Figure A.2: Average portfolio value over time (Rate: 2.5%; total inflation)

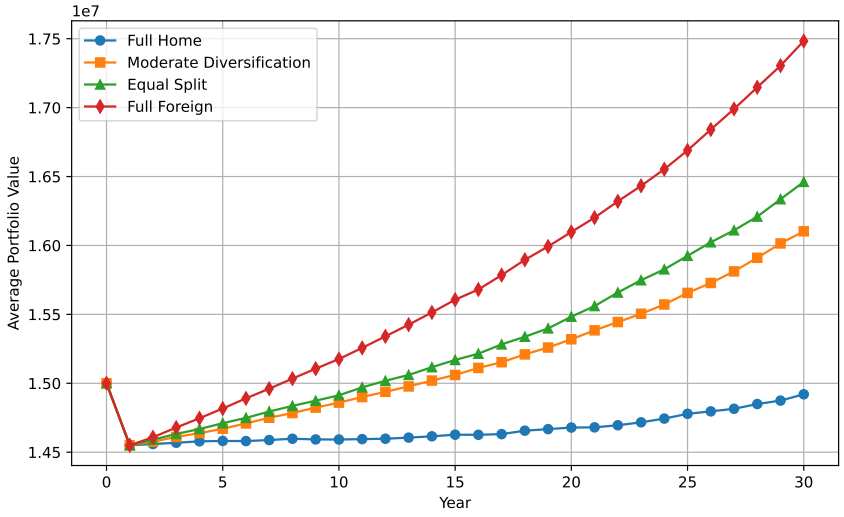


Figure A.3: Average portfolio value over time (Rate: 3%; total inflation)

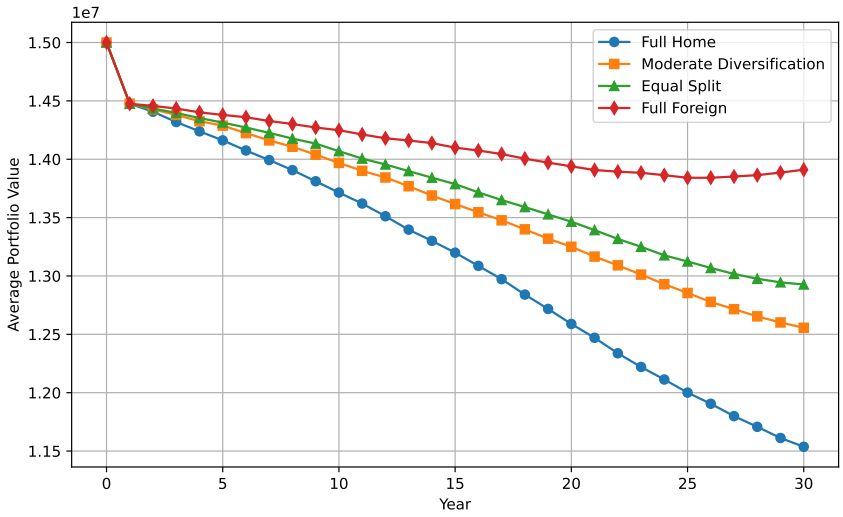


Figure A.4: Average portfolio value over time (Rate: 3.5%; total inflation)

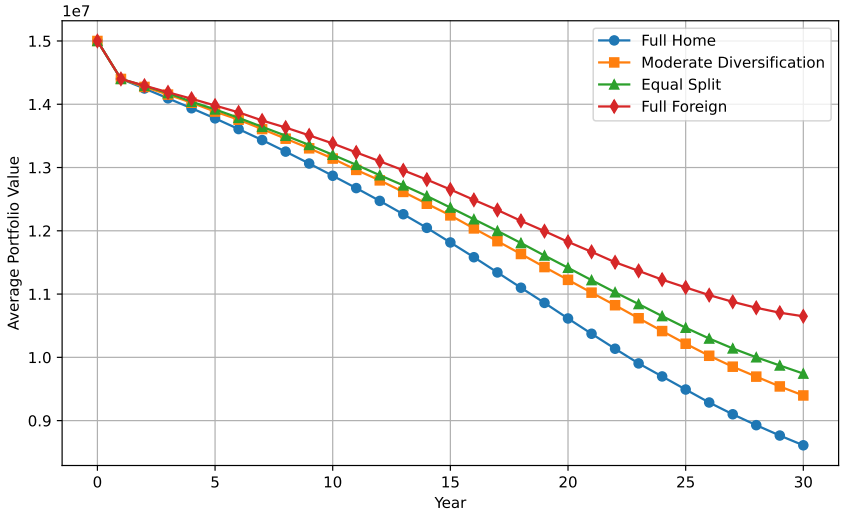


Figure A.5: Average portfolio value over time (Rate: 4%; total inflation)

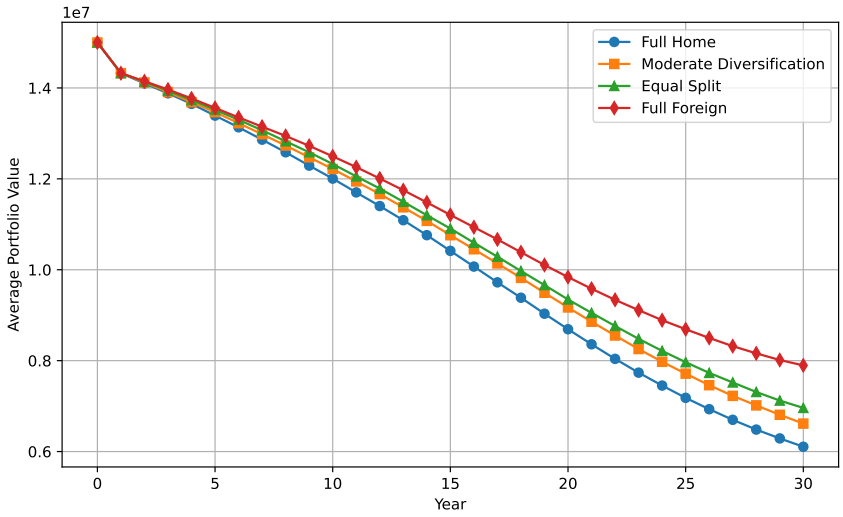


Figure A.6: Average portfolio value over time (Rate: 4.5%; total inflation)

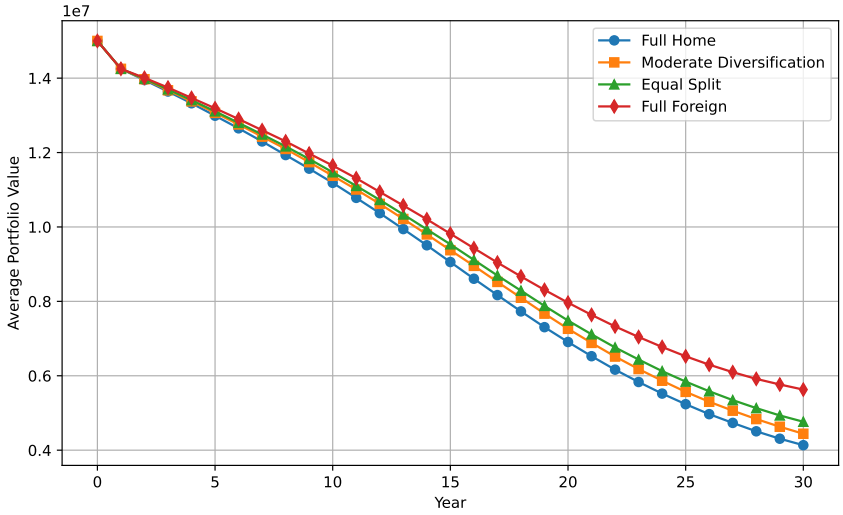


Figure A.7: Average portfolio value over time (Rate: 5%; total inflation)

A.2 Underestimated Inflation

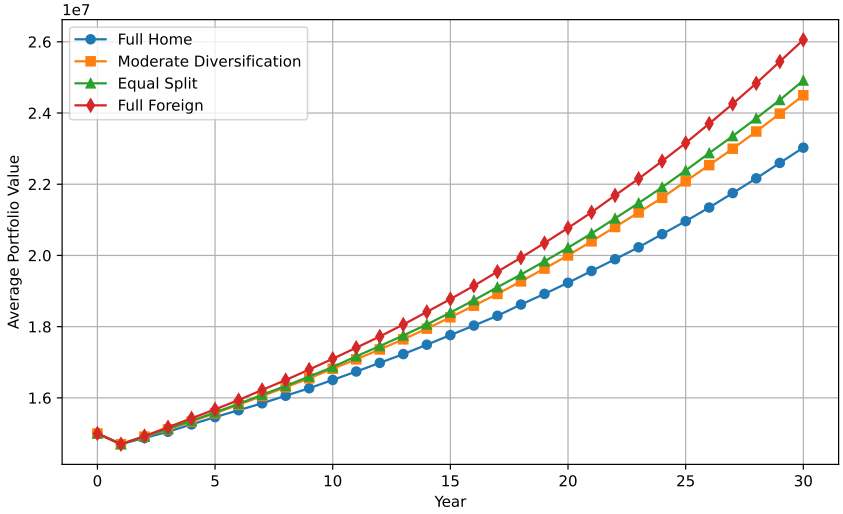


Figure A.8: Average portfolio value over time (Rate: 2%; underestimated inflation)

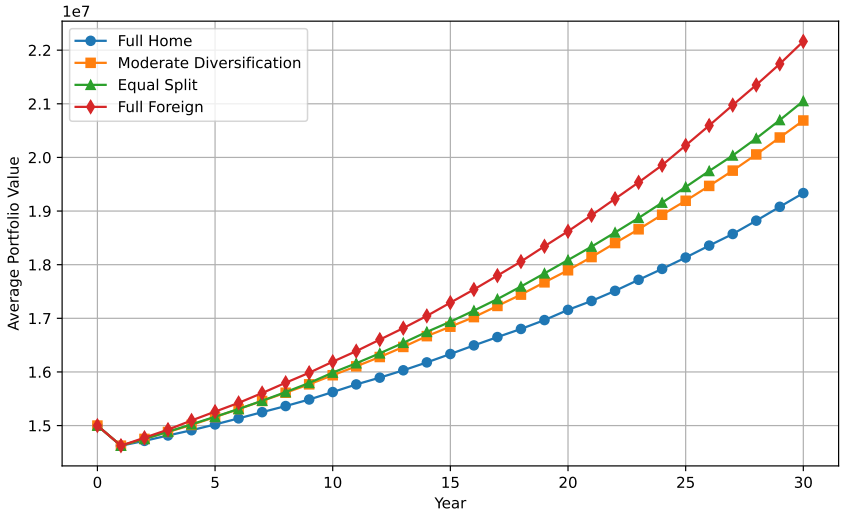


Figure A.9: Average portfolio value over time (Rate: 2.5%; underestimated inflation)

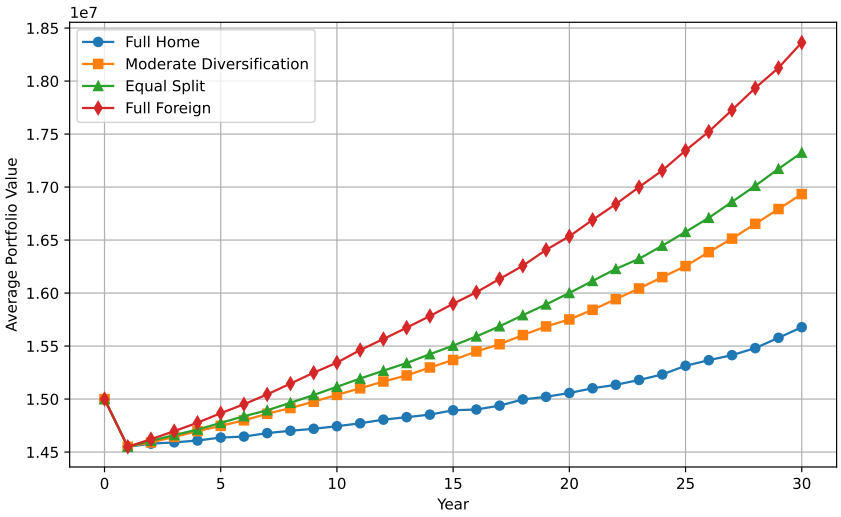


Figure A.10: Average portfolio value over time (Rate: 3%; underestimated inflation)

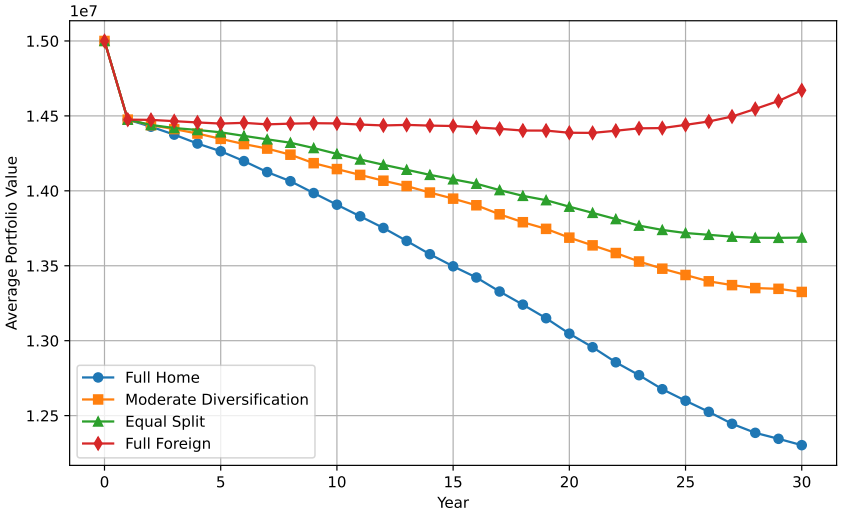


Figure A.11: Average portfolio value over time (Rate: 3%; underestimated inflation)

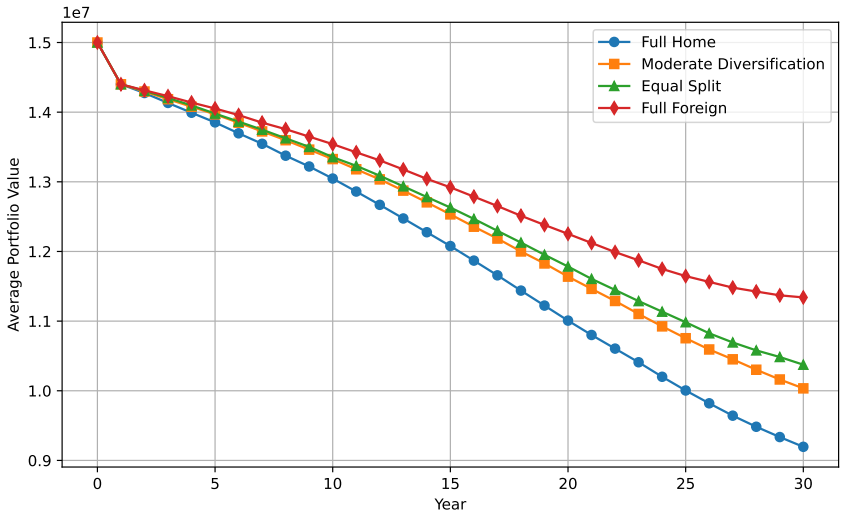


Figure A.12: Average portfolio value over time (Rate: 4%; underestimated inflation)

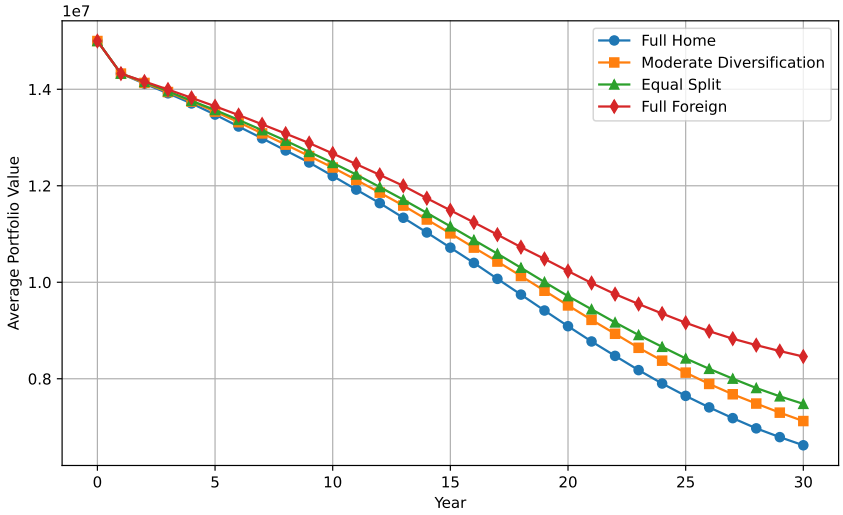


Figure A.13: Average portfolio value over time (Rate: 4.5%; underestimated inflation)

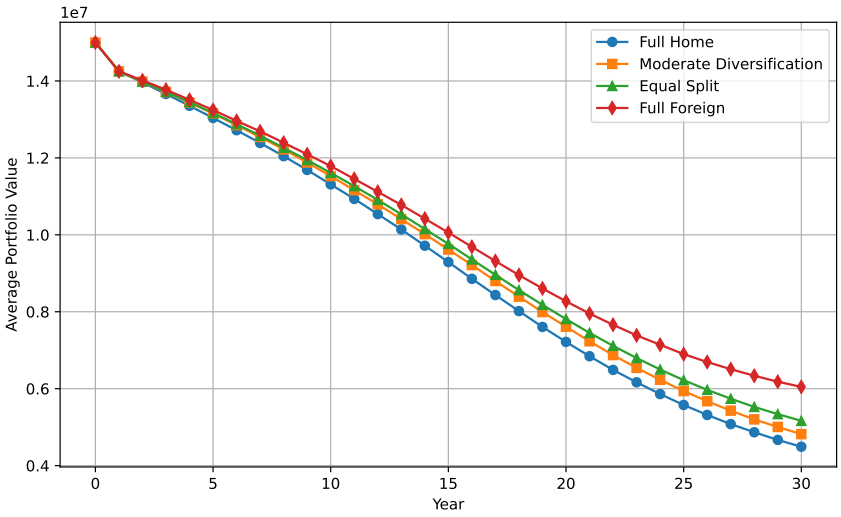


Figure A.14: Average portfolio value over time (Rate: 5%; underestimated inflation)